Tutorial: basics

Installing Kicad:
We will use a docker container image of kicad. Our container environment „apptainer“ can read the docker container format:

```
apptainer run docker://kicad/kicad:nightly
```

In the zitipool, we use a precompiled image:

```
apptainer run /shares/zitipoolhome/nca-opt/kicad_image.sif
```

The first time it will download the docker container. If you use it a second time, it will use the cached image. If the docker images changes on the docker server, apptainer will automatically update it.
In the apptainer environment, we start kicad (and detach it from the shell, since it is a GUI application):

```
apptainer> kicad &
```

In the kicad window, select „new projcet“.

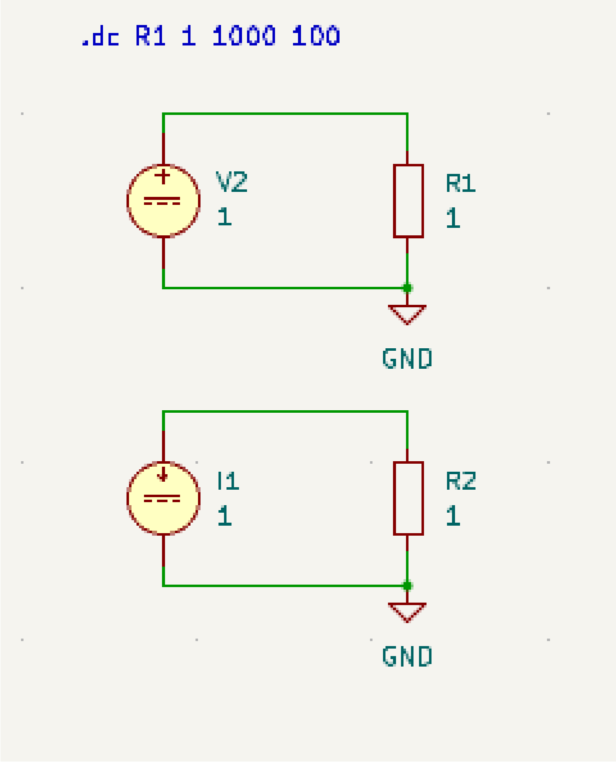
Using Windows, just download the installer and run it.

Tutorial lesson 1: basic circuits

The brain uses electricity to communicate between nerve cells. In neuromorphic computing we want to emulate the brain with electronic circuits. The following lesson will introduce the necessary basic concepts.

We always start from a closed electrical circuit, connecting at least one energy source to a number of resistors and capacitors.

KiCad lesson 1 (Kc1)
Create two elementary circuits, one using a voltage source and one a current source:



To understand voltage and current at the resistors, we use Ohm‘s law: R = U/I

Tutorial lesson 1: basic circuits – current and voltage sources – ideal sources

Q1.1:

Run the simulation and observe voltage and current at the resistors for different values of V, I and R1 and R2. You should be able to verify Ohm‘s law for the resistors. Write it down:

Hint: select DC Sweep Analysis and sweep over V2 and I1: plot voltage vs current for both circuits

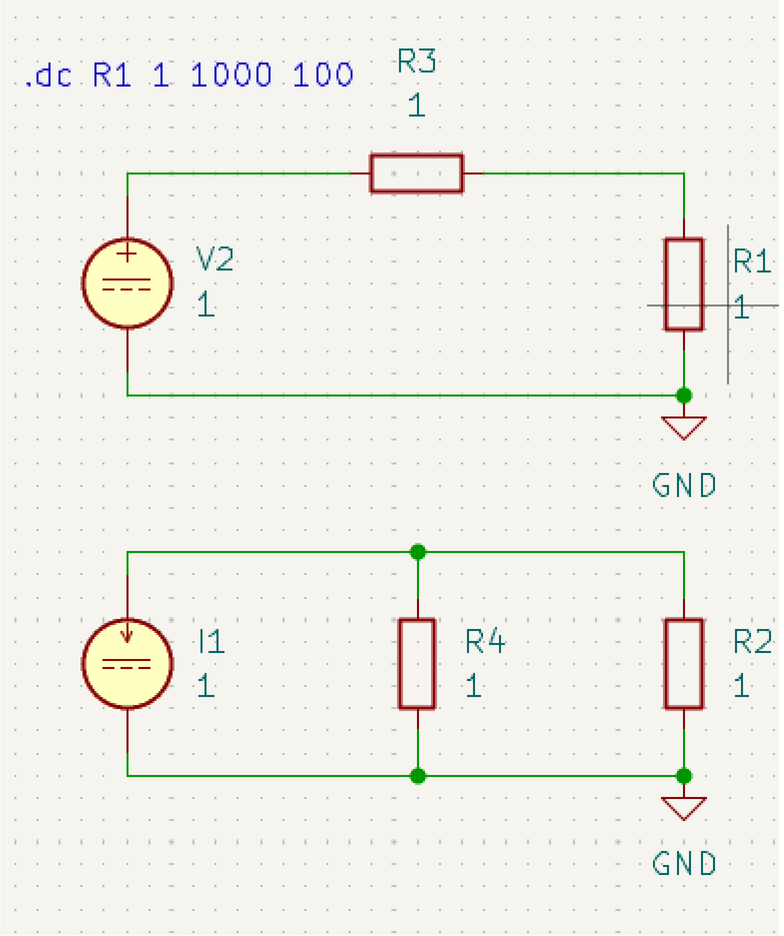
Q1.2:

Read out the values of V, I for the two sources while varying R1 and R2, respectively. How does it change? Can you deduce the equivalent resistances for the two sources?

Hint: select DC Sweep Analysis and sweep over R1 and R2: plot V(R1) and I(R2) vs V2 and I1 respectively

Tutorial lesson 1: basic circuits – current and voltage sources – real sources

KC2: Extend the schematic to include a resistor resembling the internal resistance of the source. The combination of an ideal source with a resistance is called a real source:



Tutorial lesson 1: basic circuits – Kirchoff‘s laws – calculating voltages and currents of a circuit

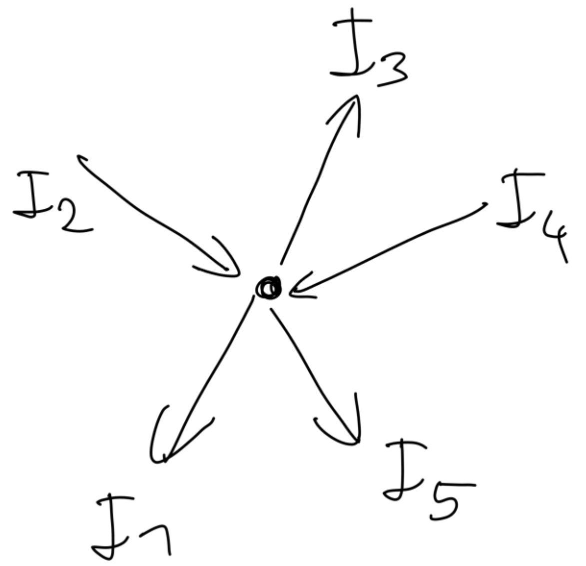
If we add more components to the circuit, we need some rules to derive the equations for all voltages and currents in the circuit -> Kirchhoff‘s and Ohm‘s laws (including ideal sources)

Use them to solve the next question:
Q2.1: What is the VDC to IDC ratio that always produces the same voltage at R1 resp. R2 for a given Rs? Is it possible to deduce the nature of the source if you only know V and I?

Gustav Robert Kirchhoff, 1824-1887, was a professor in Heidelberg from 1854 to 1876, the KIP Logo is a reference to his work

Kirchhoff's Current Law: KCL

The sum of all currents entering and leaving any node of a circuit is always zero.



Convention: currents leaving the node are positiv

$I_1 - I_2 + I_3 - I_4 + I_5 = 0$

Underlying law of physics: charge conservation

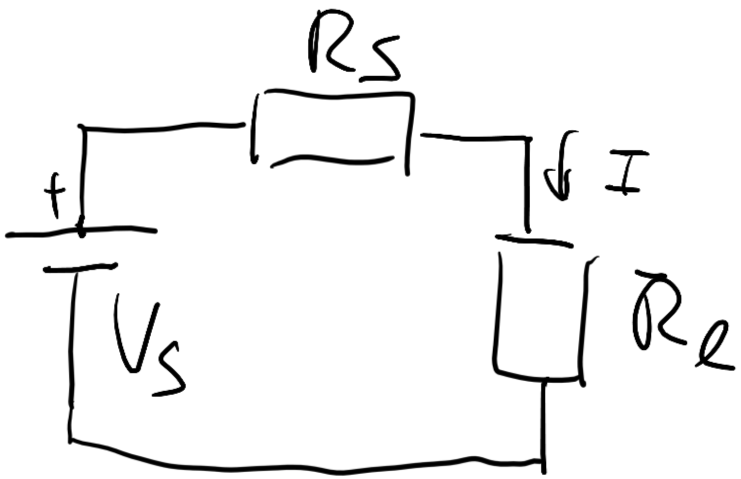
Tutorial lesson 1: basic circuits – Kirchhoff‘s laws - KVL

Kirchhoff‘s Voltage Law, KVL

The sum of all voltages along any closed loop in an electrical circuit is zero.

Example:

$-V_s+I*R_s+I*R_l=0$



Conventions:
1. Positive voltage drop along the direction of current flow in a resistor
2. Negative voltage drop along a voltage source from – to +

Note: KCL is used to argue that the current I is the same everywhere in the circuit.

Underlying law of physics: conservation of energy

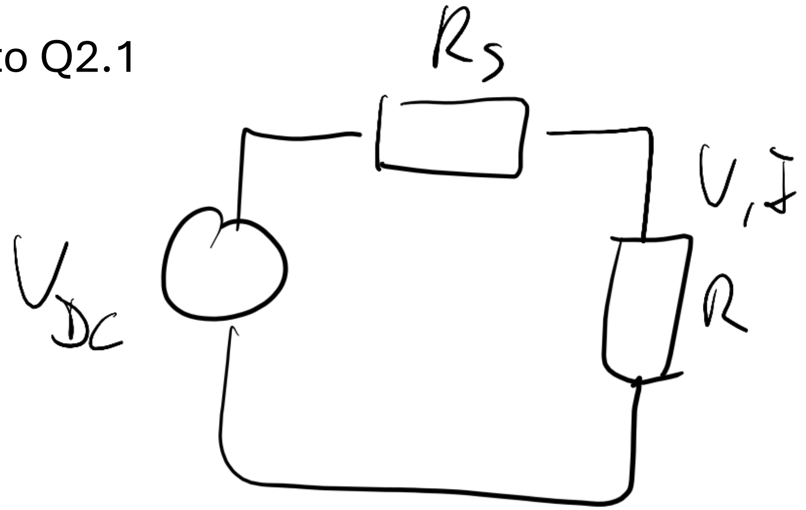
What does „voltage drop“ mean? Voltage = energy per charge

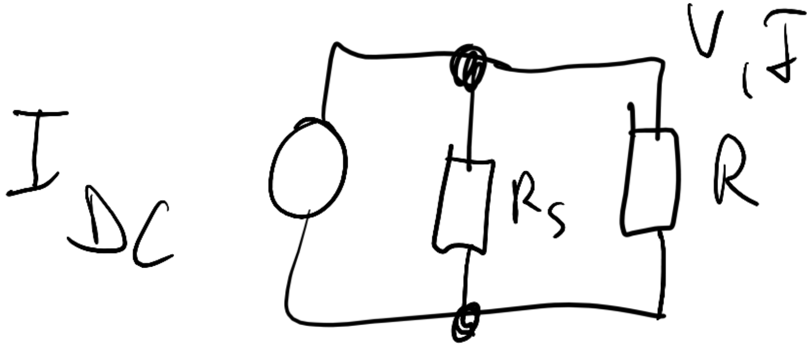
Voltage drop (Volt):
Energy (Joule) the charge (Coloumb) looses by passing the resistor.
Where does the energy go -> heat!

Thus, to keep a constant current (charge per time) flowing in a circuit loop, no energy must be lost. All losses have to be compensated by energy sources (i.e. batteries) → Sum of all voltages = 0

Tutorial lesson 1: basic circuits – Kirchoff's laws – calculating voltages and currents of a circuit

Hint to Q2.1





$$V = V_{DC} - R_S \cdot I \quad (1) \quad \text{KVL}$$

$$I = I_{DC} - \frac{V}{R_S} \quad (2) \quad \text{KCL}$$

(1) in (2):

$$I = I_{DC} - \frac{V_{DC}}{R_S} + I$$

$$\frac{V_{DC}}{I_{DC}} = R_S$$

Tutorial lesson 1: basic circuits – capacitance

Time plays an important role in the brain. The circuits we have looked at so far are not dependent on time. Time dependency enters the picture if we allow the storage of charge (-> cell membrane). The electrical component that allows us to store charge is the capacitor:

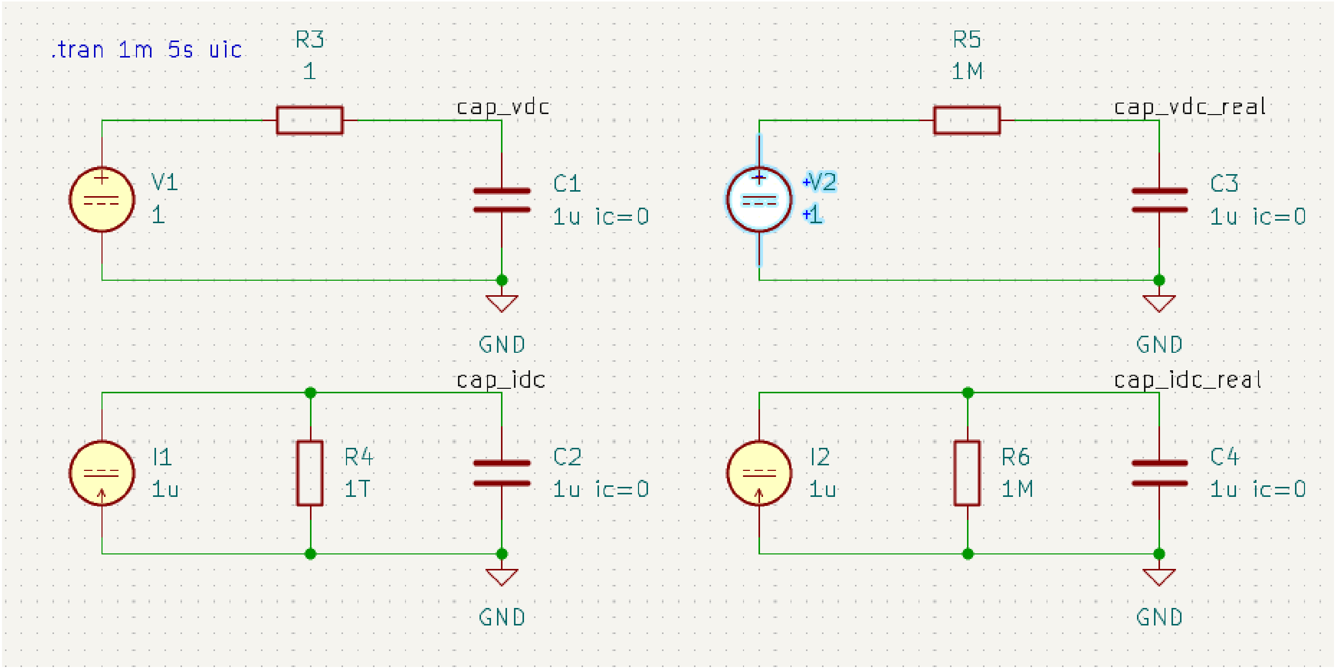
Vcapacitor = Qstored / capacitance   capacitance C[Farad] = const

Qstored is the sum of all charge that has flown on the capacitor from t=-inf to now. With I(t)=dQ(t)/dt one gets:

Q=Integral(I(t)dt) -> V(t) = Int(I(t)dt) / C

Tutorial lesson 1: basic circuits – capacitance – charging and discharging curves

KC3: replace the resistors R1 and R2 by capacitors.



For the simulation to operate properly we have to specify the charge on the capacitor at the beginning of our simulation (we do not integrate from t=-inf, but some arbitrary time t=0). We set this charge to zero by adding „ic=0“ behind the value of the capacitance (just separated by space). Ic is short for „initial condition“.

To simulate the charging and discharging of capacitances, we have to simulate over time: select „transient analysis“ in the simulator and check the „use initial conditions“ box.

Tutorial lesson 1: basic circuits – capacitance – charging and discharging curves

Q3.1: What mathematical function describes V(t) if the capacitor is connected to an ideal current source?

Hint: look at Vcap_idc(t) in the lower left schematic

Q3.2: Compare Vcap_idc_real(t) with Vcap_vdc_real(t) to inspect the case of a real source

Hint: use the values of R5 and R6 given in the schematic first, then vary R5 and R6. Compare with the result from Q2.1 in both cases.

technical hints:
In your apptainer shell enter:

```
cp -R /shares/zitipoolhome/nca-opt/bci ~
cd ~/bci
kicad &
```

Open project in kicad will show you all KC lesson files.

The bci directory can also be downloaded in moodle.

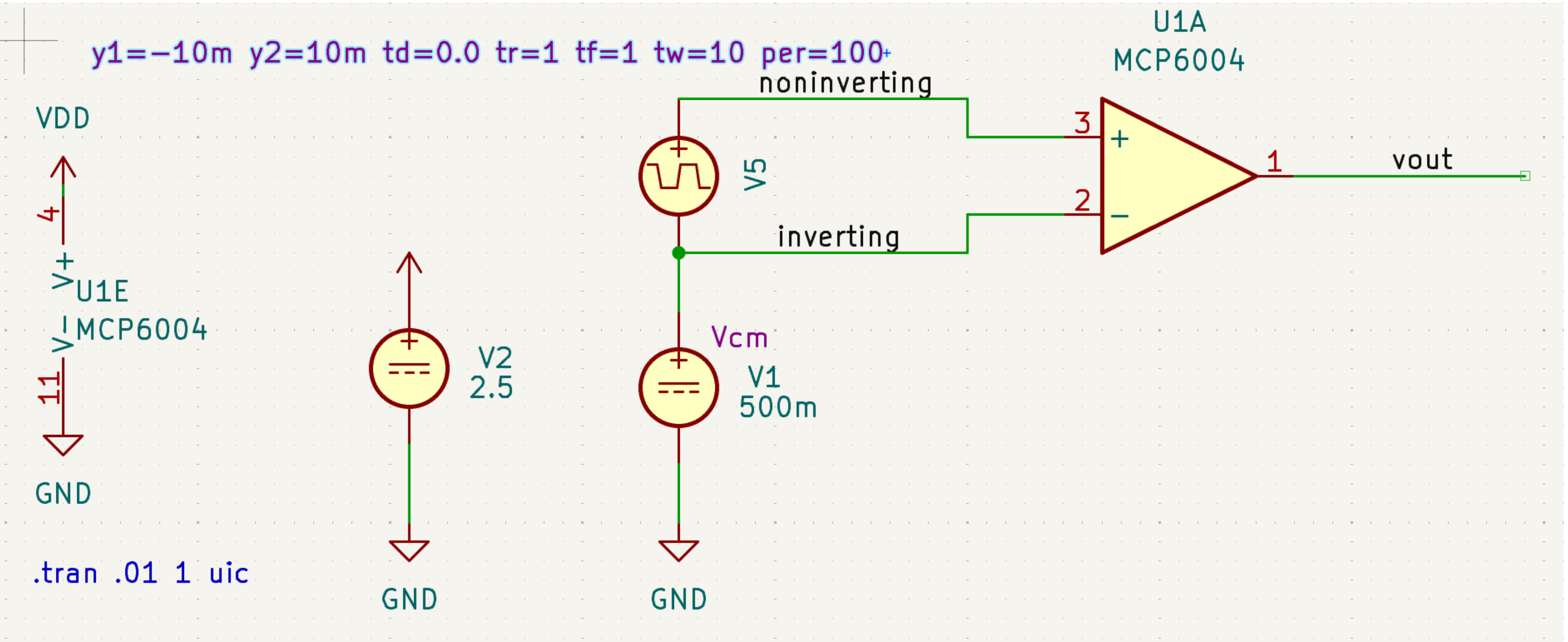
Tutorial lesson 2: complex circuits – amplifiers – operational amplifier basics

Neurons are more than linear integrators. Their voltage-gates ion channels create complex dynamics.

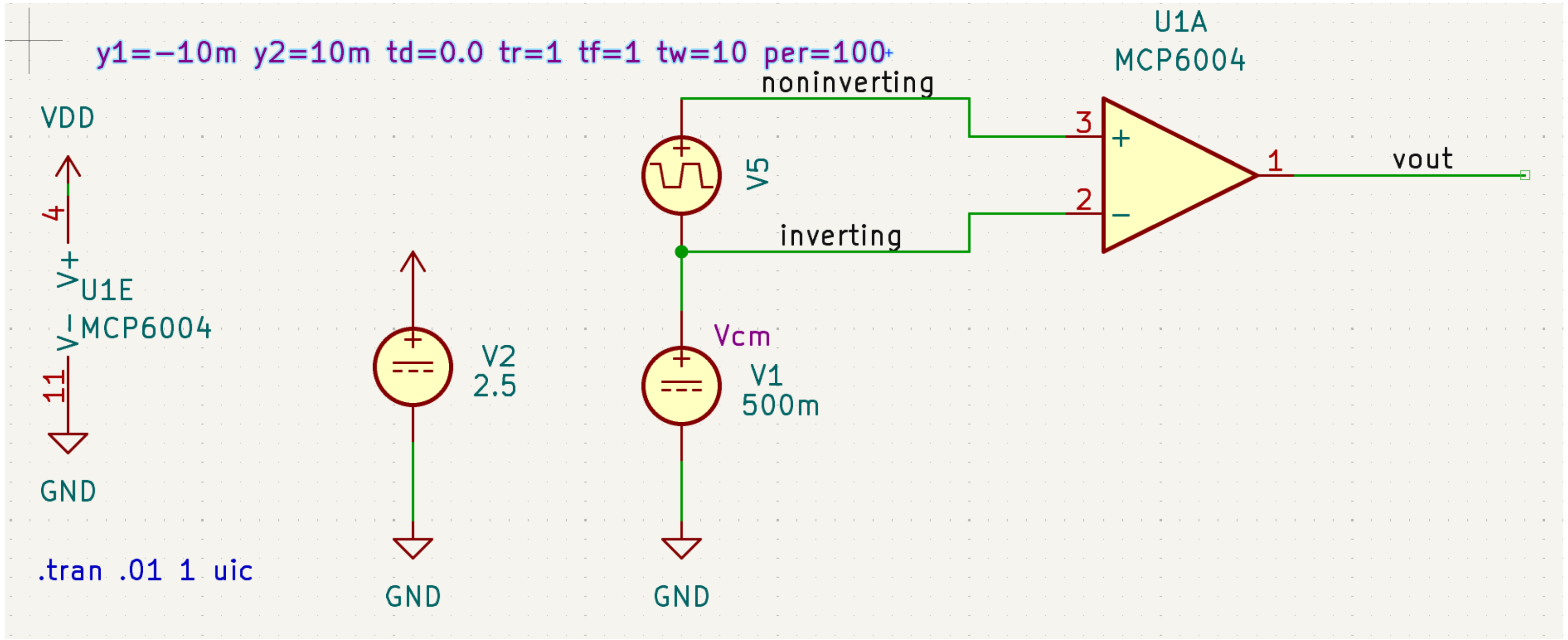
To replicate neural dynamics with analog electronic circuits, we need a way to change the voltage of our sources during operation, using *controlled sources* .

A voltage controlled voltage source is the operational amplifier, short OA. Our neuron will use the MC6004 integrated operational amplifier.

KC4: MC6004 is an operational amplifier (OA).
It hast an output V_{out} and two inputs $V_{noninverting}$ (short +) and $V_{inverting}$ (-).



Tutorial lesson 2: complex circuits – amplifiers – operational amplifier basics

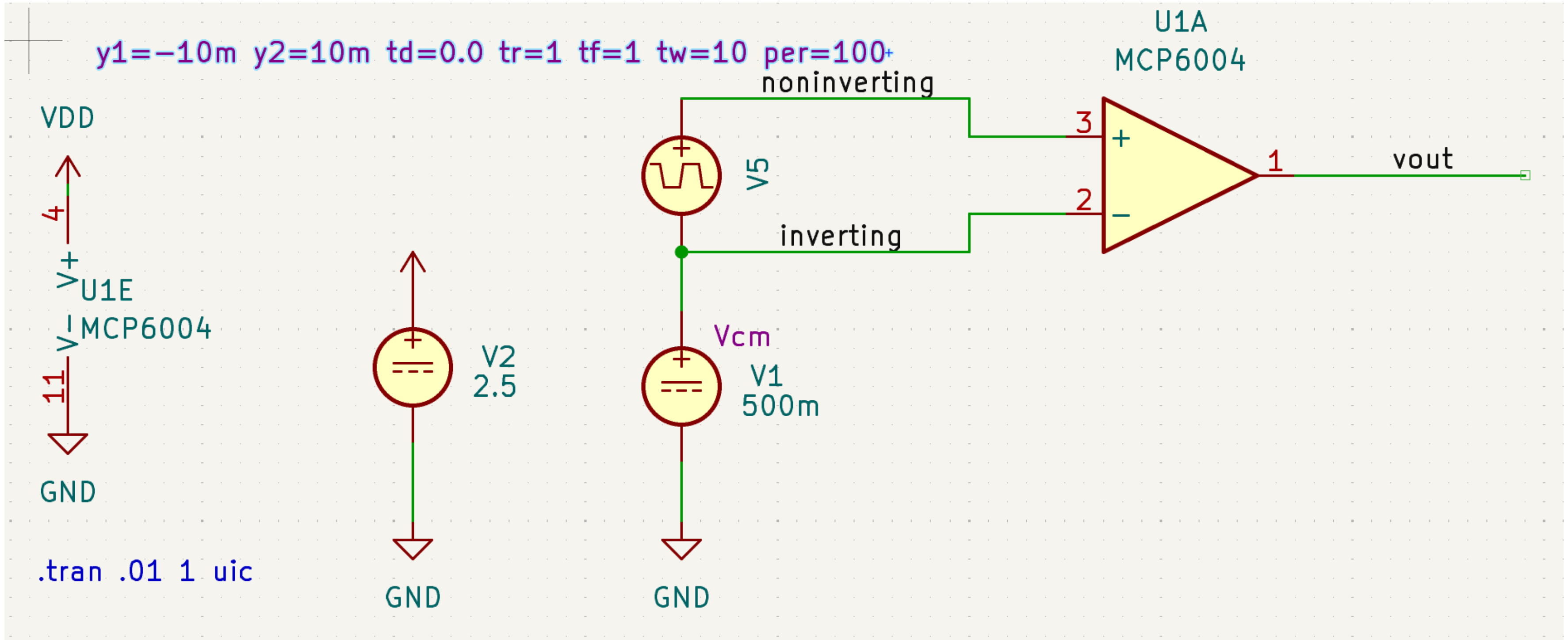


Q4.1: Verify that it amplifies $V_{in} = V_{noninverting} - V_{inverting}$, i.e. $V_{out}(V_{in}) = A * (V_{noninveriting} - V_{inverting})$.
Determine A from the simulation.

Hint: A real amplifier has small deviations from the ideal behaviour : $V_{out} = A*(V_{in} + V_{err})$.
Also, its output voltage is limited to the interval between GND and VDD.
Since A is very high and V_{err} is unknown, we can not measure A exactly: the amplifier is mostly stuck at GND or VDD.
We can see that it takes only a very small change in V_{in} for the amplifier to transistion from GND to VDD, this gives us a lower bound on A.

Q4.2: Does it amplifiy V_{cm}?
Hint: Vary V_{cm} in the simulation and observe whether the transistion of V_{out} from GND to VDD changes.

Tutorial lesson 2: complex circuits – amplifiers – operational amplifier basics



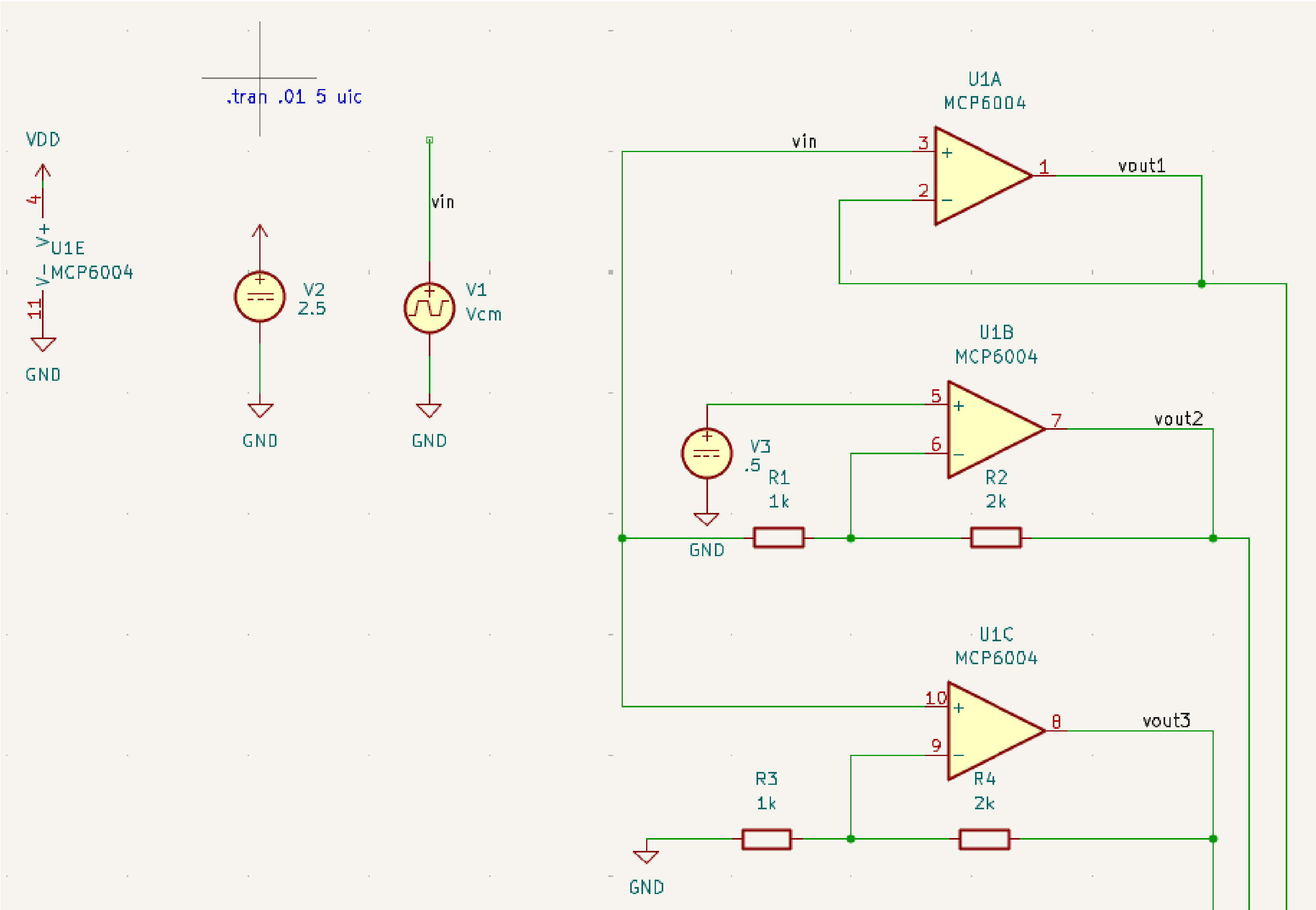
An amplifier in this configuration is said to be an open-loop amplifier. Due to the high value of A it is used to determine the sign of V_{in} independent of the amplitude. This function is termed a comparator, since it compares the voltage on both inputs and conveys the result of this comparison via ist output voltage: V_{out} =VDD if V_{in} >0 and V_{out} = GND for V_{in} <0.

Tutorial lesson 2: complex circuits – amplifiers – feedback

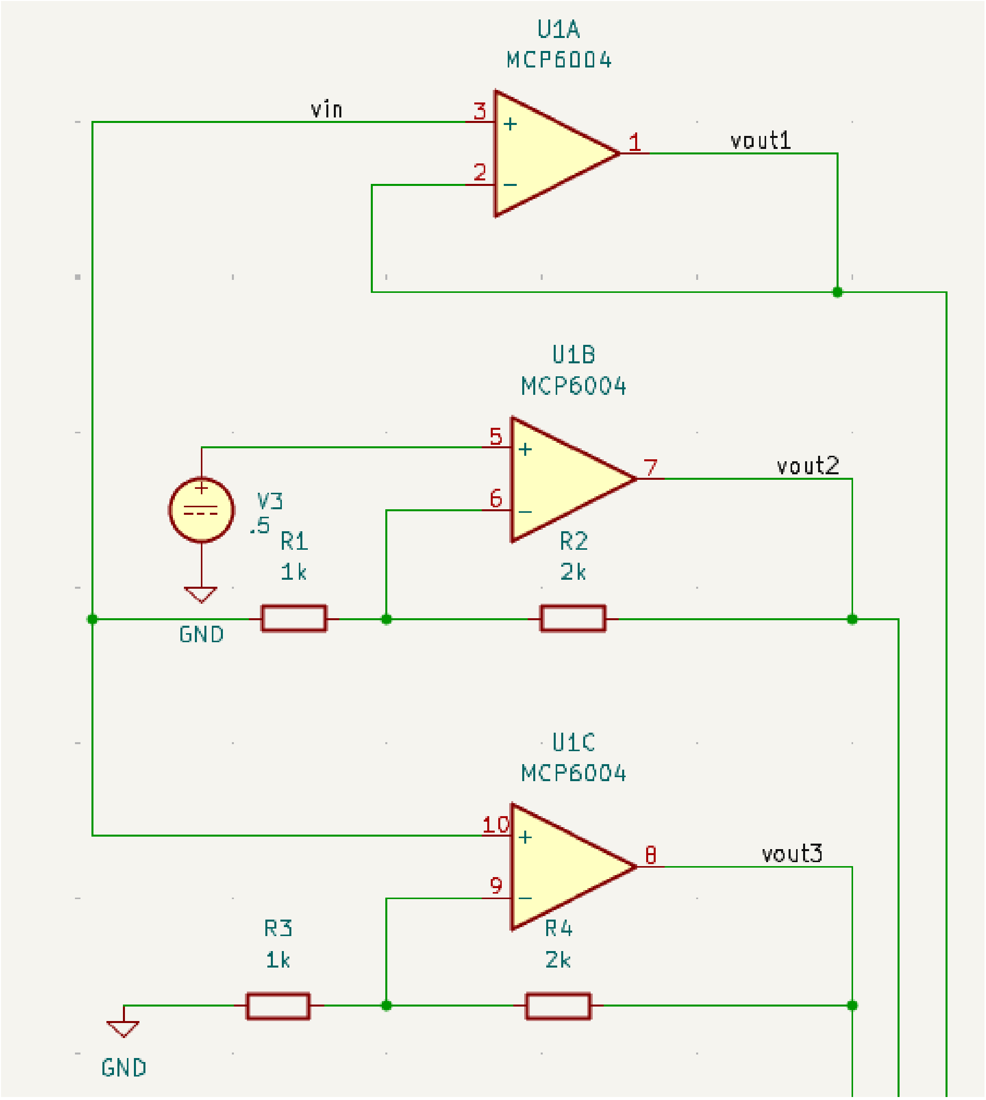
KC5: try the following configurations for the amplifier.

Q5.1:
derive the three functions $V_{out}(V_{in})$.
Treat the Op like a voltage controlled $(V_+ - V_-)$ ideal voltage source.

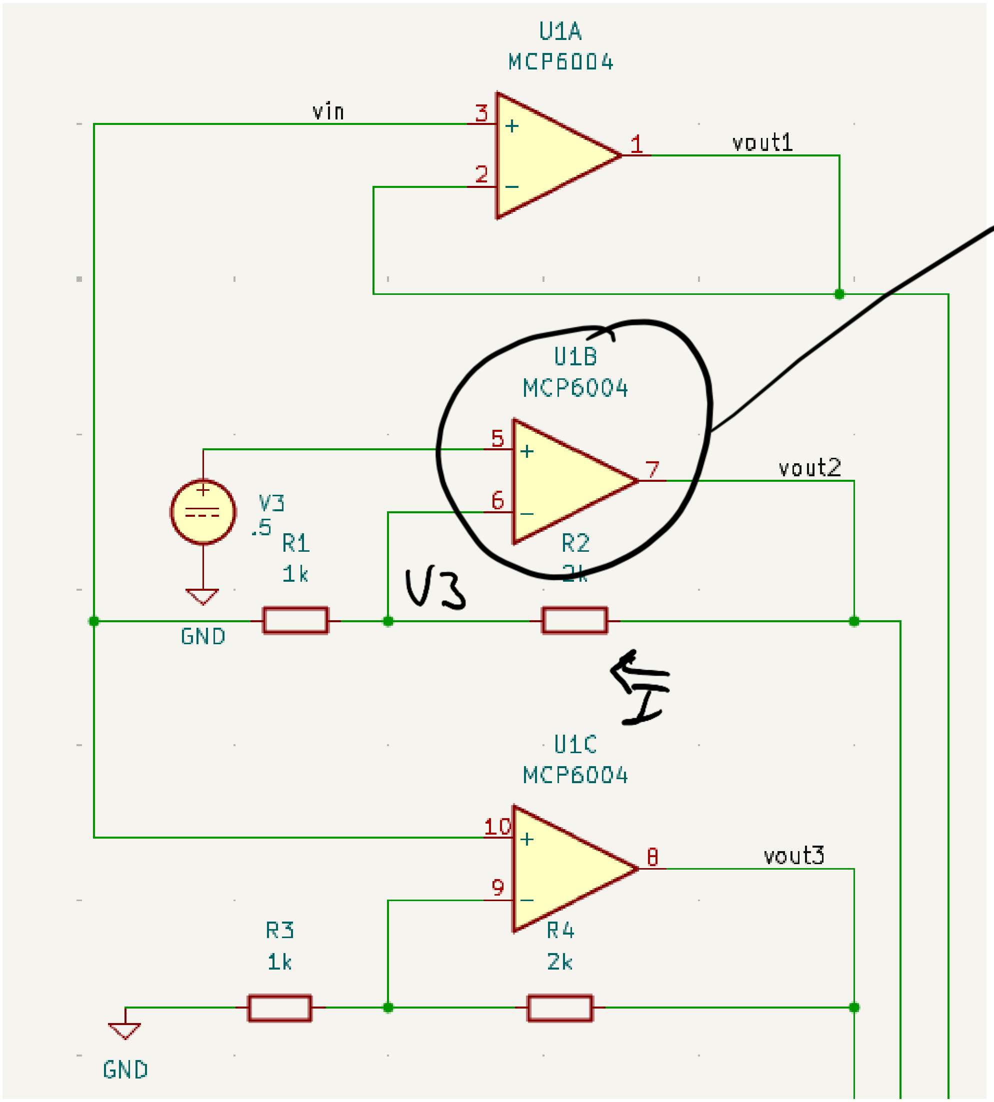
Hint:
Assume, that the amplification A of the OP is so high (~10^6) that the difference between the two inputs is nearly zero whenever the output voltage is not gnd or vdd. Use this fact to answer the question, i.e. use
$(V_+ - V_-)=0$
and calculate $V_{out}(V_{in})$ using Kirchhoffs laws.
The input resistance of the + and – pins is so high that you can assume $I_+=0$ and $I_- = 0$.



Tutorial lesson 2: complex circuits – amplifiers – feedback



Tutorial lesson 2: complex circuits – amplifiers – feedback



$$V_3 = V_{OP-}$$

$$I_{R1} = I_{R2} = I \quad \text{KCL}$$

$$V_{out2} - V_3 = I \cdot R_2 \quad \text{KVL}$$

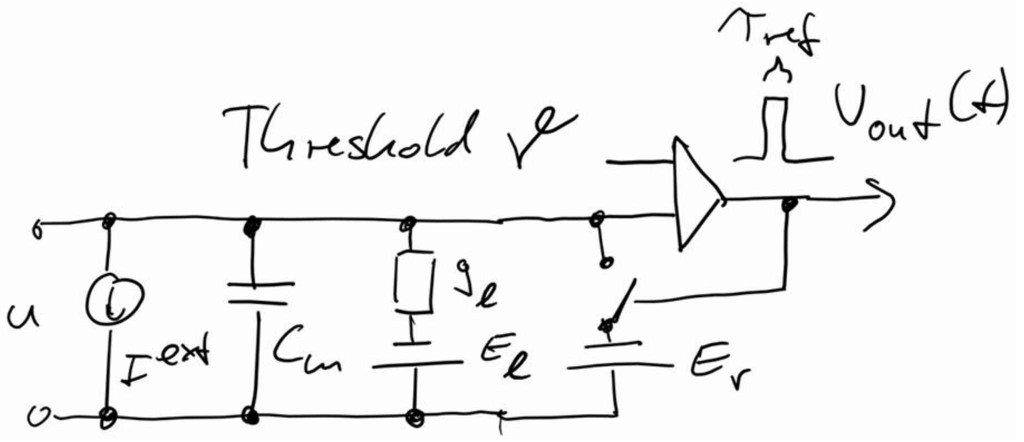
$$V_3 - V_{in} = I \cdot R_1$$

$$\frac{V_{out2} - V_3}{R_2} = \frac{V - V_{in}}{R_1}$$

$$\frac{V_{out2} - V_3}{V_3 - V_{in}} = \frac{R_2}{R_1}$$

$$\Rightarrow V_{out2} = \frac{R_2}{R_1}(V_3 - V_{in}) + V = -V_{in}\frac{R_2}{R_1} + V_3\left(1 + \frac{R_2}{R_1}\right) : \text{inverting amplifier}$$

Tutorial lesson 3: first LIF circuit - basics



The integrate and fire model allows us to build a circuit that can mimic a (simplified) biological neuron.

For the comparator we can use the operational amplifier without feedback. Due to ist high gain A its output switches from 0 to Vdd when the input voltage difference changes polarity:
$u - V_{threshold} < 0 \rightarrow V_{out} = 0$
$u - V_{threshold} > 0 \rightarrow V_{out} = Vdd$

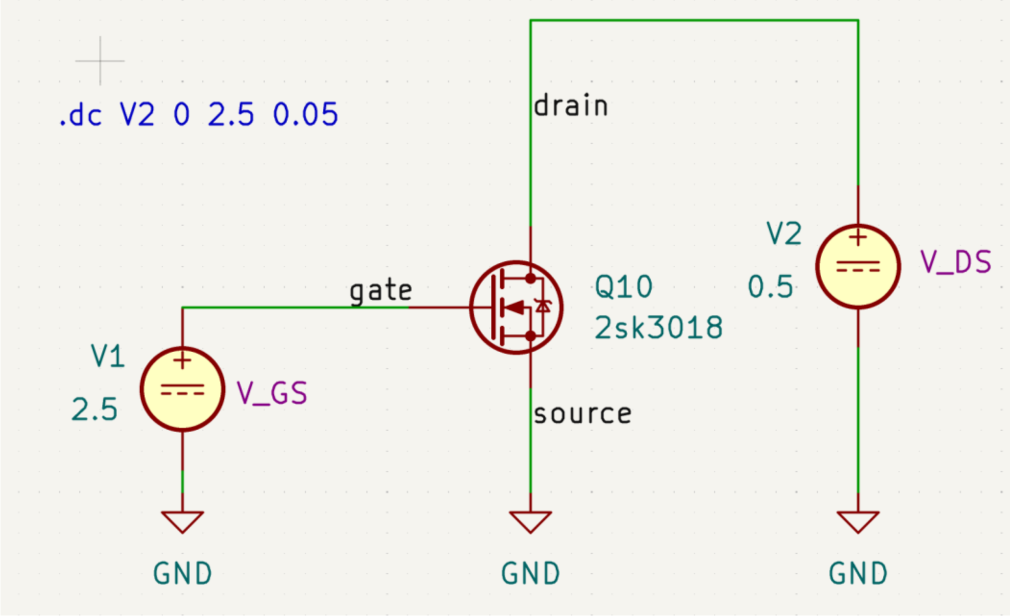
This circuit introduces a new component: a voltage controlled switch that discharges the membrane capacitance and resets u to Ereset.

We use a MOSFET, short for metal-oxide-semiconductor field-effect transistor in our circuit for this task.

Tutorial lesson 3: first LIF circuit – MOSFET

The MOSFET transistor is the basis of all our modern electronics. The amplifiers used in out neuron circuits are also build from transistors (although some are of a different kind).

A MOSFET transistor has three terminals: gate, drain and source. The voltage between gate and source V_{GS} controls the current that flows from drain to source, I_{DS}. If V_{GS} is zero, no current can flow, i.e. $I_{DS} = 0$.
The gate has a near infinite resistance, i.e. no current flows into the gate
→ the transistor is (an approximation of) a voltage-controlled current source



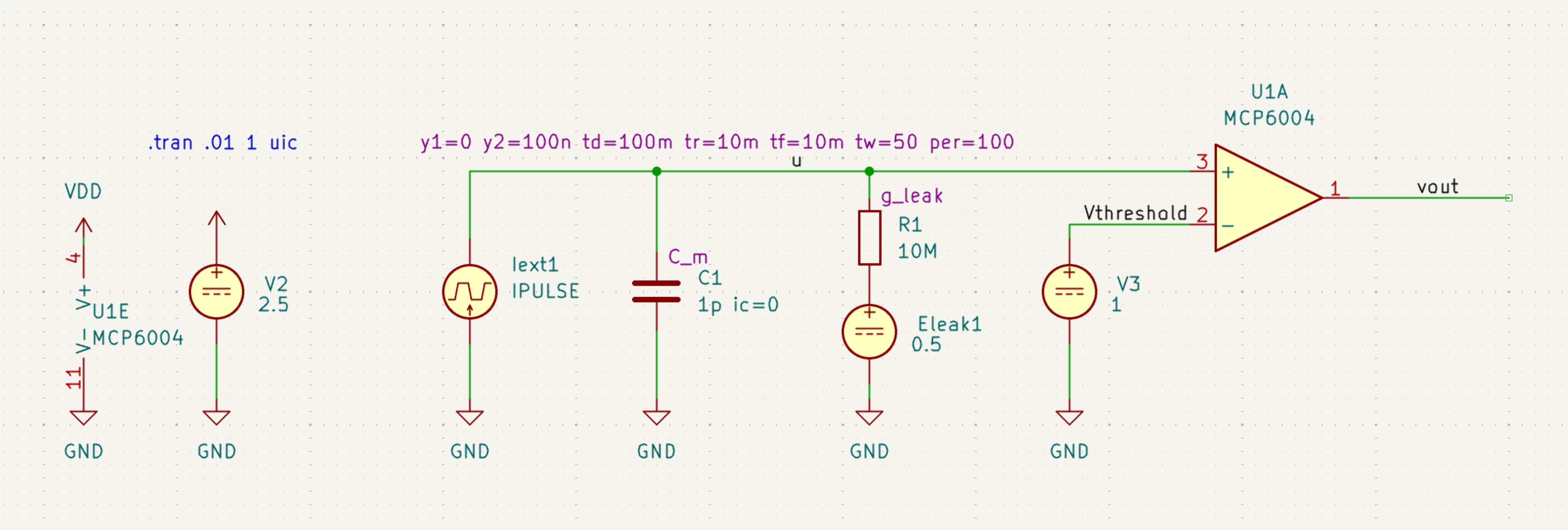
In our LIF circuit it is used only as a switch: V_{GS} is either zero or V_{DD}. The current through the transistor is either zero or high. In this configuration the transistor approximates a switch.

KC6: MOSFET
Q6.1 Simulate the output characteristics I_DS(V_DS) for different values of V_GS (0,1,2,2.5 Volts). What is the lowest resistance you can achieve limiting V_DS to a maximum of 2.5 Volts?

Tutorial lesson 3: first LIF circuit

KC7: A simple LIF circuit

In your bic3 directory using the apptainer shell: `kicad 3_first_lif.kicad_pro &`

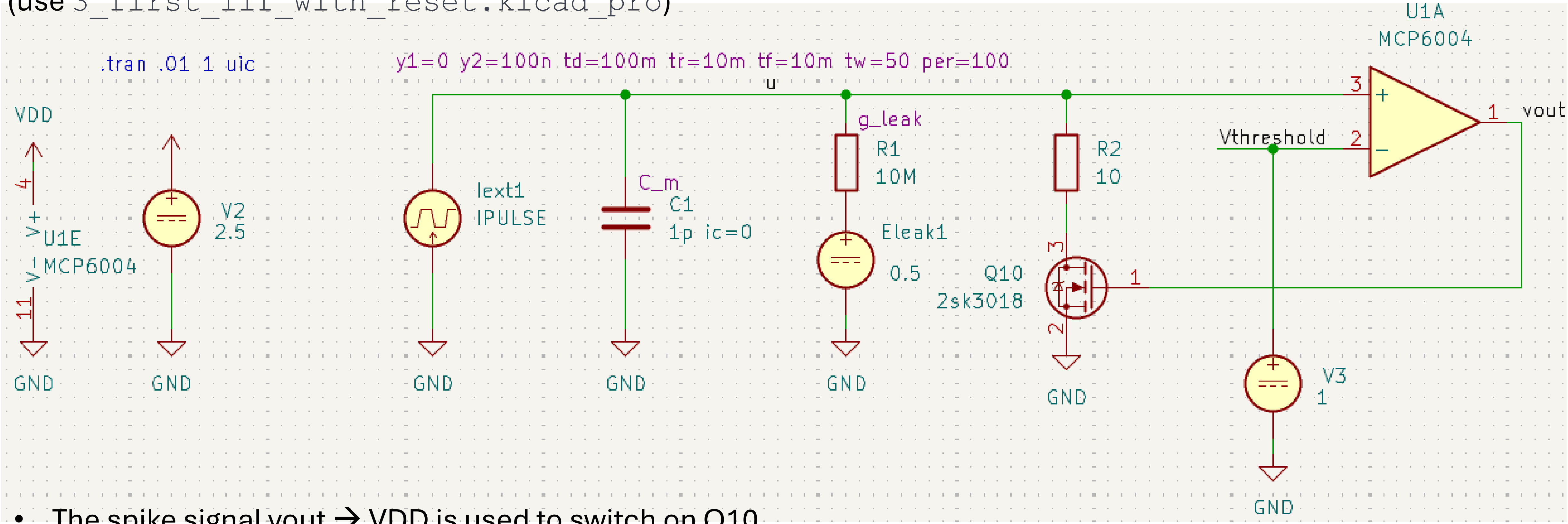


This circuit is missing the reset-mechanism, so after the membrane voltage u has crossed Vthreshold, the ouput stays at VDD (2.5V) forever.
To fit the model to the parametes of our circuit, the voltages have all been shifted and scaled to fit the range of our power supply Vdd=2.5 Volts.

Q7.1 Observe the dependency of u(Eleak) and vout(Eleak). Use the tuned value option for the Eleak source in the schematic. Change also the input current.

Tutorial lesson 3: first LIF circuit – adding a reset

KC8: We complete the LIF by inserting a reset circuit using a transistor as a switch:
(use `3_first_lif_with_reset.kicad_pro`)

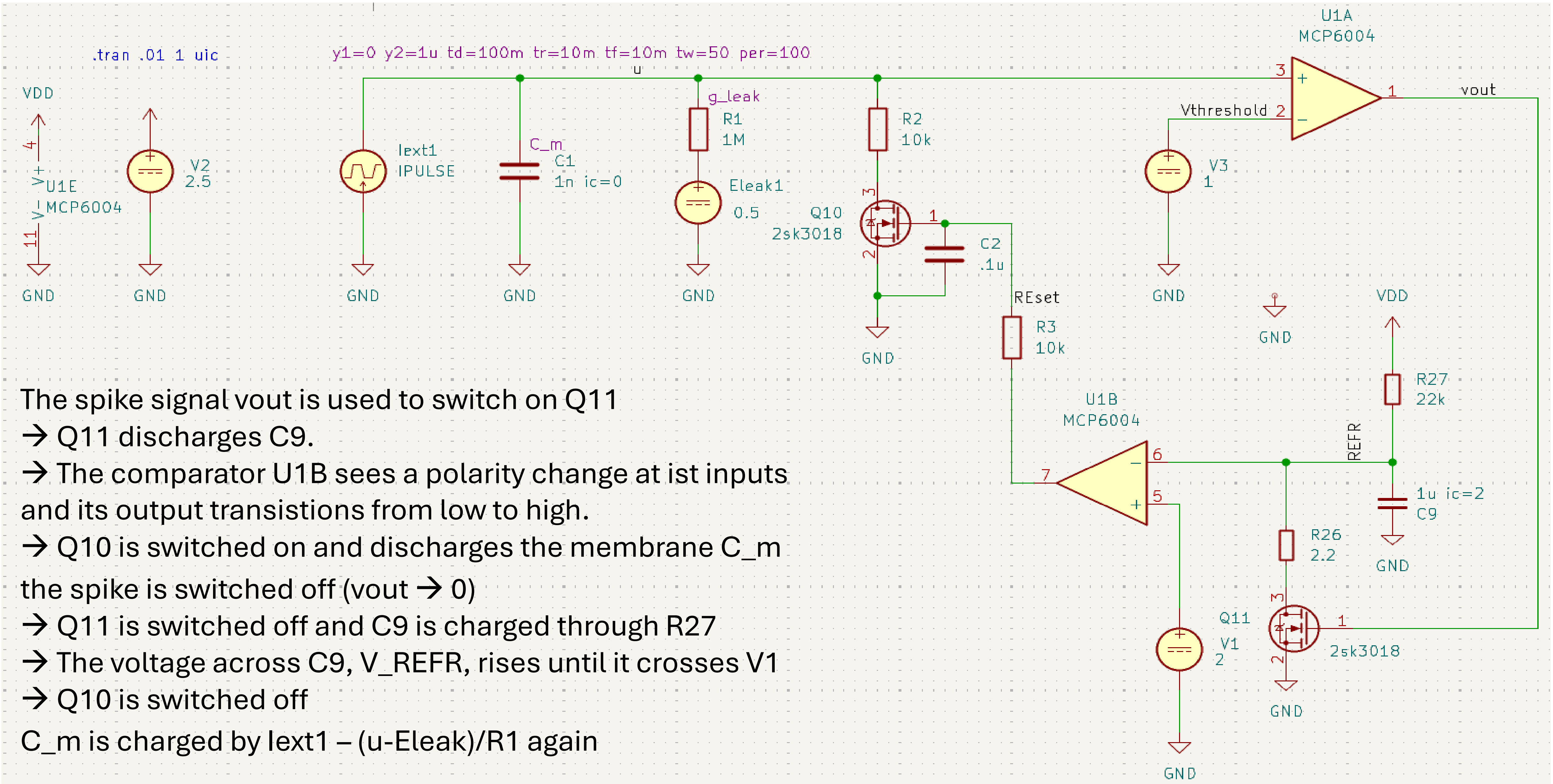


- The spike signal vout → VDD is used to switch on Q10
  → Q10 discharges C_m.
  → The comparator U1A sees a polarity change at ist inputs and changes vout → 0
  → Q10 is switched off
- C_m is charged by Iext1 – (u-Eleak)/R1 again

Q8.1 Observe the vout frequency in dependency of the Iext pulse height. Verify that it is a Type I neuron.

Tutorial lesson 4: first LIF circuit – adding a true refractory period

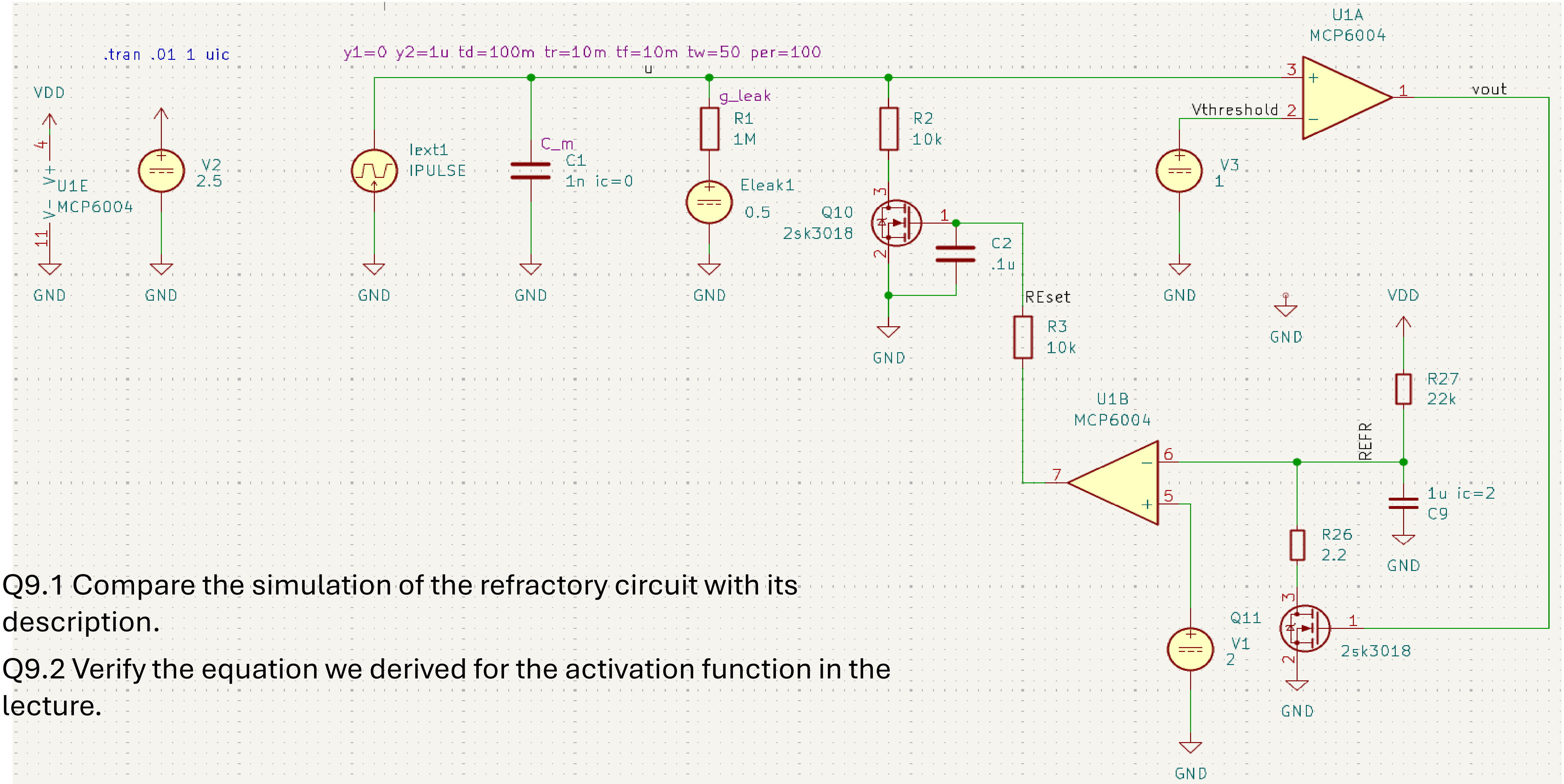
KC9: To create a longer and adjustable refractory period, we have to add a refractory circuit:
(use `3_first_lif_with_refractory.kicad_pro`)



- The spike signal vout is used to switch on Q11
  → Q11 discharges C9.
  → The comparator U1B sees a polarity change at ist inputs and its output transitions from low to high.
  → Q10 is switched on and discharges the membrane C_m
- the spike is switched off (vout → 0)
  → Q11 is switched off and C9 is charged through R27
  → The voltage across C9, V_REFR, rises until it crosses V1
  → Q10 is switched off
- C_m is charged by Iext1 – (u-Eleak)/R1 again

Tutorial lesson 4: first LIF circuit – adding a true refractory period

KC9: To create a longer and adjustable refractory period, we have to add a refractory circuit:
(use `3_first_lif_with_refractory.kicad_pro`)



Q9.1 Compare the simulation of the refractory circuit with its description.

Q9.2 Verify the equation we derived for the activation function in the lecture.

Tutorial lesson 5: first LIF circuit – adder circuit for synaptic currents

KC9: the three output voltages are combined by resistors at the noninverting input of the amplifier U1D.

The amplifier U1D follows the same equation as derived in Q5.1 for U1C. One has just to replace V3 with the combination of Vout1 to 3 generated by R7 – R10.
With KCL we can write:

$$I_{R10} = I_{R7} + I_{R8} + I_{R9}$$

$$V_+ = I_{R10} \cdot R_{10}$$

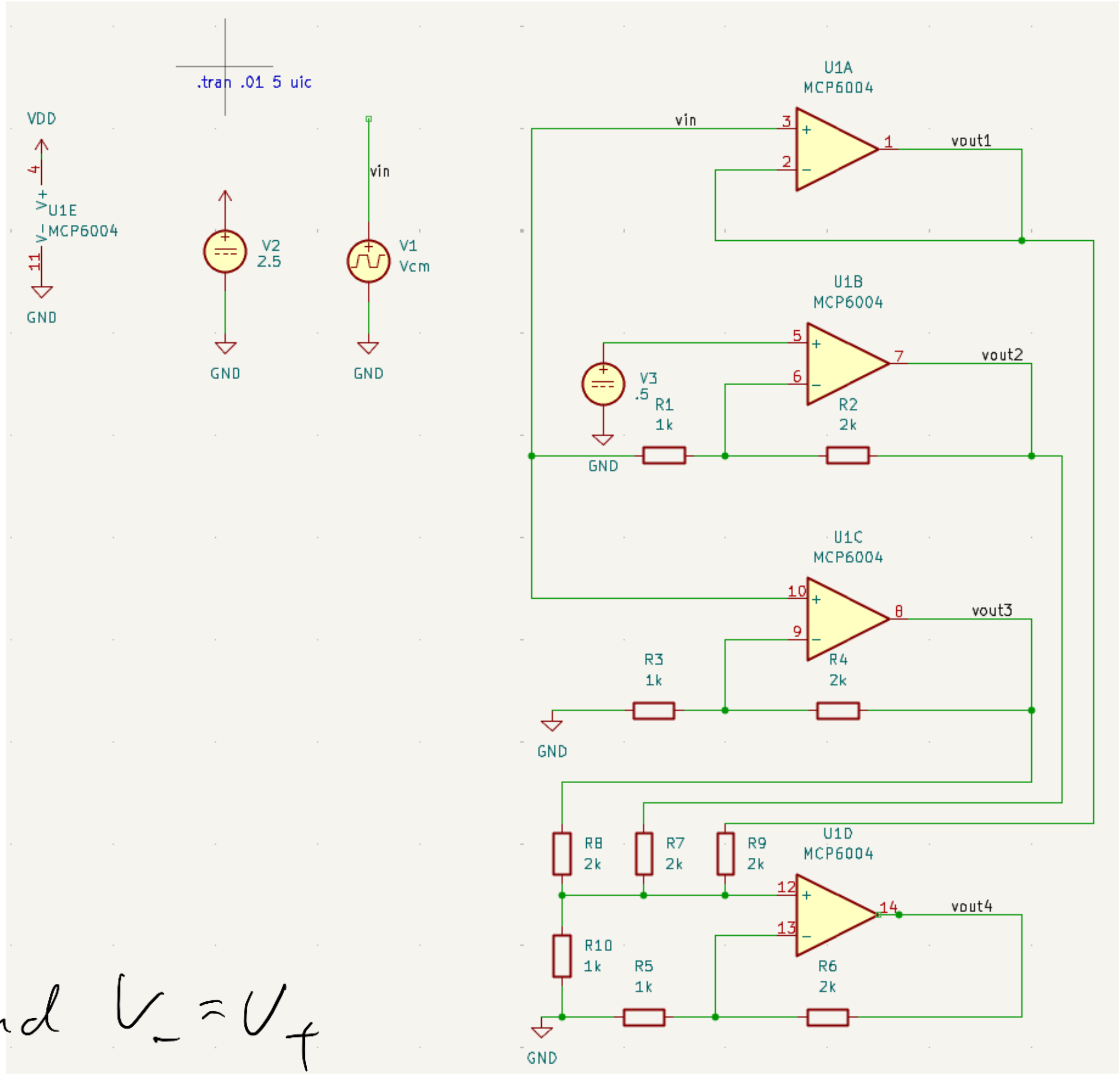
$$I_{R10} = \frac{V_{out1}}{R_9} + \frac{V_{out2}}{R_7} + \frac{V_{out3}}{R_8}$$

$$R_7 = R_8 = R_9 = R$$

$$V_+ = \frac{R_{10}}{R}\left(V_{out1} + V_{out2} + V_{out3}\right)$$

Using Q5.1: $\frac{V_{out4}}{V_-} = \frac{R_5 + R_6}{R_5}$ and $V_- = V_+$

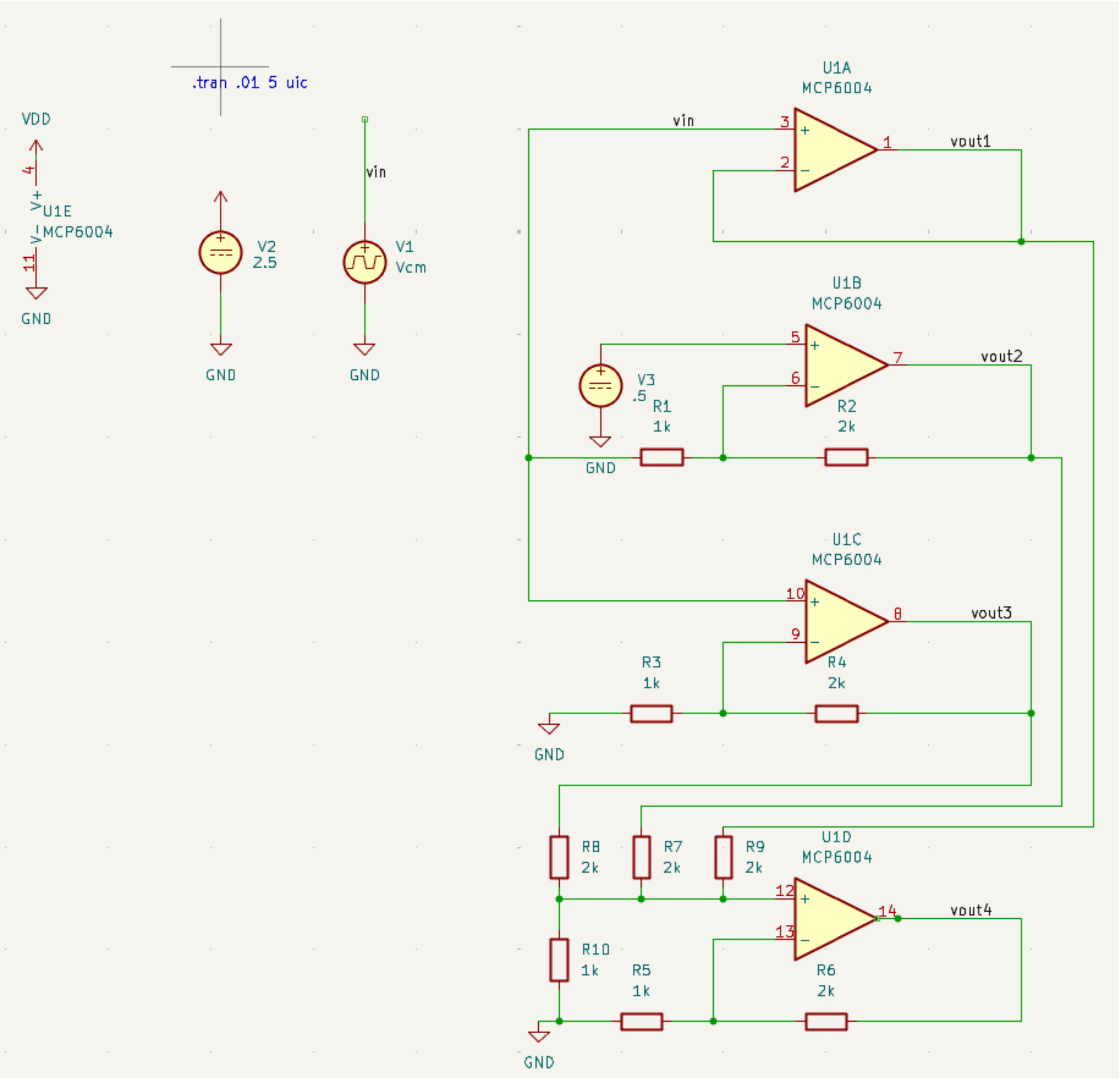
$$\Rightarrow V_{out4} = \left(1 + \frac{R_6}{R_5}\right) V_+$$



Tutorial lesson 5: first LIF circuit – adder circuit for synaptic currents

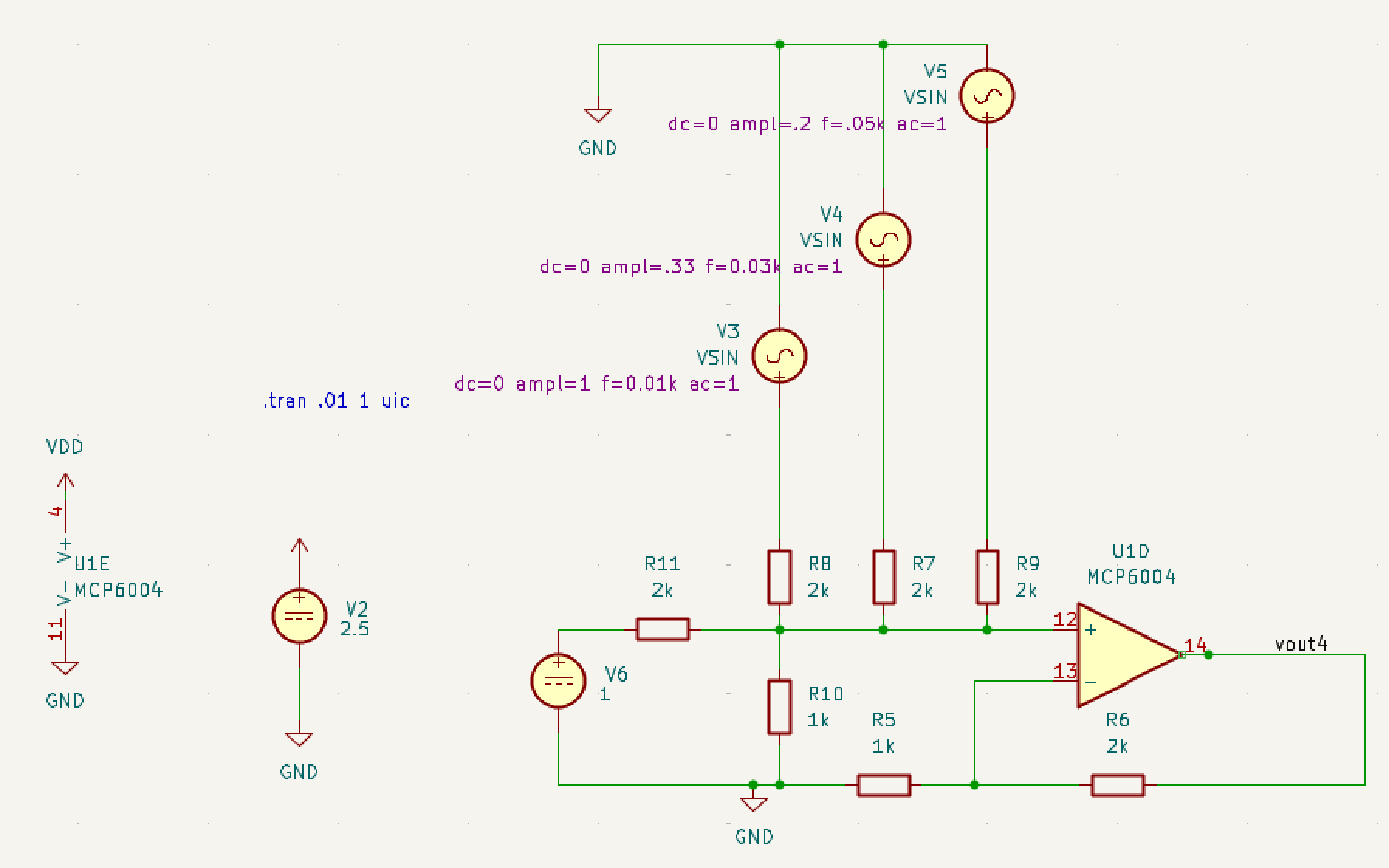
V_{out4} is the sum of the three input voltages times a constant. This scheme can be extended to many more inputs.
In our LIF circuit we can use it to add the currents from different synapses on the membrane of our neuron!
Instead of a complicated multi-bit digital adder circuit the summation is carried out by using one of the basic laws of physics: charge conservation.
Calculating in such a way is often called „physical computing".

Q9.1: Simulate the circuit and compare with the solution calculated above.
Q9.2: Derive the function $V_{out4}(V_{out1,2,3})$ of the circuit in the case of $R_7 \mathrel{!}= R_8 \mathrel{!}= R_9$ and verify in simulation.
Q9.3: Do you have an idea how to substract an input voltage?



Tutorial lesson 5: first LIF circuit – adding time-continuous signals

KC10: Since the input spikes to a neuron arrive asynchronously the neuron has to add the synaptic inputs continuously over time.
We simulate such a scenario by replacing three OPs by VSIN sources as you see in the picture. VSIN is an ideal voltage source that produces a sinusoidal voltage instead of a DC voltage.
The input signals have been chosen in a way to represent the first three components of the Fourier-series of a pulse wave (https://de.wikipedia.org/wiki/Rechteckschwingung).



Q10.1: Investigate in simulation that the result from KC9 still holds.
Q10.2: Modify the circuit in a way that you get the same Vout4 while using VSIN sources with identical amplitudes (ampl=1 for all sources) and verify in simulation.